MMML CLI Cheat Sheet

This cheat sheet explains the `mmml` command-line interface at the level needed for running and teaching the tutorial workflow. Use:

```
mmml --help
mmml <command> --help
```

for the authoritative option list on your installed version.

Mental Model

The CLI has five main jobs:

- Build molecular systems: `make-res`, `make-box`, `run-pycharmm`
- Generate QM/ML data: `pyscf-dft`, `pyscf-mp2`, `normal-mode-sample`, `pyscf-evaluate`, `fix-and-split`
- Train and evaluate models: `physnet-*`, `ef-*`, `train`, `evaluate`, `extract-checkpoint-metrics`
- Run dynamics: `run`, `md-system`, `physnet-md`, `ef-md`
- Inspect, convert, and visualize: `validate`, `xml2npz`, `unwrap-traj`, `gui`, `kernel-fit`

Common file conventions:

- Geometry samples: NPZ with `R`, `Z`, `N`
- E/F/D training data: NPZ with `R`, `Z`, `N`, `E`, `F`, `Dxyz`
- ESP data: NPZ with `esp`, `esp_grid` or split `grids_esp_*.npz`
- PhysNet checkpoints: Orbax run directories or portable JSON checkpoints
- EF checkpoints: usually `params.json` plus config in an output directory

Tutorial One-Liners

These are the commands people usually need during the water-dimer tutorial:

```
cd examples/mmml_tutorial/cli_water

bash 01_make_res_cli.sh
bash 02_make_box_cli.sh
bash 04_pyscf_dft_cli_full.sh
bash 06_normal_mode_sample_cli.sh
bash 07_pyscf_evaluate_cli.sh
bash 08_fix_and_split_cli.sh
bash 10_physnet_dcmnet_train_cli.sh
```

Fast DCMNet smoke test with prepared `out/splits_dimer/` files:

```
JAX_PLATFORMS=cpu uv run python -m mmml.cli.misc.train_joint \
  --train-efd out/splits_dimer/energies_forces_dipoles_train.npz \
  --train-esp out/splits_dimer/grids_esp_train.npz \
  --valid-efd out/splits_dimer/energies_forces_dipoles_valid.npz \
  --valid-esp out/splits_dimer/grids_esp_valid.npz \
  --use-repo-physnet-params \
  --epochs 1 \
  --batch-size 1 \
  --name smoke_repo_physnet \
  --ckpt-dir /tmp/mmml_ckpts \
  --plot-freq 0
```

System-Building Commands

`mmml make-res`

Generate residue/topology files with PyCHARMM/CGENFF.

```
mmml make-res --res TIP3 --skip-energy-show
mmml make-res --res CYBZ
```

Key options:

- `--res`: residue name.
- `--skip-energy-show`: skip a slow CHARMM validation call on clusters.

Typical outputs:

- `pdb/initial.pdb`
- `psf/initial.psf`
- `xyz/initial.xyz`
- CHARMM topology/parameter files

Requires CHARMM/PyCHARMM.

`mmml make-box`

Pack molecules into a periodic box.

```
mmml make-box --res TIP3 --n 2 --side_length 25.0
mmml make-box --res CYBZ --n 50 --side_length 25.0 --solvent TIP3 --density 1.0
```

Key options:
- `--res`: residue to pack.
- `--n`: number of molecules.
- `--side_length`: cubic box side length in Angstrom.
- `--solvent`, `--density`: optional solvated setup.

Typical output:
- `pdb/init-packmol.pdb`

Requires CHARMM/PyCHARMM and PackMol.

`mmml run-pycharmm`

Run pure CHARMM heating/equilibration, without ML.

```
mmml run-pycharmm --pdbfile pdb/init-packmol.pdb --cell 25.0
```

Use this for a classical baseline before MM/ML or ML-only simulations.

Typical outputs:
- `pdb/heat.pdb`, `pdb/equi.pdb`
- `dcd/heat.dcd`, `dcd/equi.dcd`
- `res/heat.res`, `res/equi.res`

QM Data Generation

`mmml pyscf-dft`

Run GPU-accelerated DFT calculations.

```
mmml pyscf-dft --mol xyz/initial.xyz --energy --gradient --output out/04_results
mmml pyscf-dft --mol xyz/initial.xyz --energy --gradient --hessian --harmonic --output
out/04_results
```

Key options:
- `--mol`: XYZ file or inline molecule string.
- `--energy`, `--gradient`, `--hessian`, `--harmonic`, `--thermo`: requested calculations.
- `--output`: output base path; writes `.npz` and `.h5`.

Typical outputs:
- `out/04_results.npz` with ML-friendly keys.
- `out/04_results.h5` with full arrays, including harmonic data when requested.

Requires PySCF/gpu4pyscf/CuPy.

`mmml pyscf-mp2`

Run GPU-accelerated MP2 calculations.

```
mmml pyscf-mp2 --mol water.xyz --energy --gradient --output out/mp2_results
```

Key options:
- `--mol`: XYZ file or inline molecule string.
- `--basis`: basis set, default `def2-SVP`.
- `--spin`, `--charge`: electronic state.
- `--energy`, `--gradient`: requested calculations.

Use MP2 as a post-HF reference path, not as the default tutorial path.

`mmml normal-mode-sample`

Sample structures from harmonic normal modes.

```
mmml normal-mode-sample -i out/04_results.h5 -o out/06_sampled.npz --amplitude 0.1 --max-samples
1000
mmml normal-mode-sample -i out/04_results.h5 -o out/06_sampled.npz --amplitudes 0.05 0.1 0.2 --
include-equilibrium
```

Key options:

- `-i`, `--input`: HDF5 file from `pyscf-dft --harmonic`.
- `-o`, `--output`: sampled geometry NPZ.
- `--amplitude` or `--amplitudes`: displacement magnitudes in Angstrom.
- `--freq-min`: skip low-frequency modes.
- `--include-equilibrium`: include the reference geometry.
- `--samples-per-mode`: 1 or 2 for + or +/- displacements.
- `--max-samples`: cap total structures.

Typical output:

- NPZ with `R`, `Z`, `N`.

`mmml pyscf-evaluate`

Evaluate many geometries in one PySCF process.

```
mmml pyscf-evaluate -i out/06_sampled.npz -o out/07_evaluated.npz --esp
mmml pyscf-evaluate -i traj.npz -o out_efield.npz --EF --efield=0,0,0.01
mmml pyscf-evaluate -i traj.npz -o noisy.npz --add-random-noise 0.1 --seed 1
```

Key options:

- `-i`, `--input`: geometry NPZ with `R`, `Z`, `N`.
- `-o`, `--output`: labeled NPZ.
- `--basis`, `--xc`, `--spin`, `--charge`: QM settings.
- `--esp`: compute electrostatic potential grid.
- `--EF`: include electric-field calculations.
- `--efield Ex,Ey,Ez`: fixed field in atomic units.
- `--efield-sigma`: random field scale.
- `--add-random-noise`: add coordinate noise before evaluation.
- `--no-energy`, `--no-gradient`, `--no-dipole`: skip targets.

Typical output:

- NPZ with `R`, `Z`, `N`, `E`, `F`, `Dxyz`
- plus `esp`, `esp_grid` when `--esp` is set
- plus `Ef` / e-field keys when electric fields are enabled

`mmml verify-esp-alignment`

Check that ESP grids and molecular coordinates are aligned.

```
mmml verify-esp-alignment out/07_evaluated.npz
```

Use this when ESP errors look suspicious or when converting external data.

Data Preparation and Validation

`mmml fix-and-split`

Convert raw QM units and create train/valid/test splits.

```
mmml fix-and-split --efd out/07_evaluated.npz --output-dir out/splits
mmml fix-and-split --efd out/07_evaluated.npz --output-dir out/splits --atomic-ref pbe0/def2-tzvp
mmml fix-and-split --efd train.npz md_evaluated.npz --output-dir out/splits_extended
```

Key options:

- `--efd`: one or more NPZ files with E/F/D data.
- `--grid`: optional separate ESP grid NPZ.
- `--output-dir`, `-o`: split output directory.

- `--train-frac`, `--valid-frac`, `--test-frac`: split ratios.
- `--seed`: reproducible shuffle.
- `--atomic-ref`: subtract per-atom reference energies.
- `--flip-forces`: use if `F` stores gradients instead of forces.
- `--energy-scale`, `--force-scale`, `--dipole-scale`, `--esp-scale`: last-resort correction factors.
- `--n-grid-points`, `--esp-max-abs-kcal-mol`, `--min-dist-to-atoms`: ESP point filtering.
- `--skip-validation`: faster but less safe.

Typical outputs:
- `energies_forces_dipoles_train.npz`
- `energies_forces_dipoles_valid.npz`
- `energies_forces_dipoles_test.npz`
- `grids_esp_train.npz`
- `grids_esp_valid.npz`
- `grids_esp_test.npz`

`mmml validate`

Validate NPZ files against the expected MMML schema.

```
mmml validate out/splits/energies_forces_dipoles_train.npz
mmml validate out/splits/*.npz
```

Use this after external conversion or manual data editing.

`mmml xml2npz`

Convert Molpro XML files into MMML NPZ format.

```
mmml xml2npz input.xml -o output.npz
mmml xml2npz inputs/*.xml -o dataset.npz --validate
```

Use this for legacy Molpro-generated datasets.

PhysNet and DCMNet

`mmml physnet-md`

Run MD using a trained PhysNet checkpoint.

```
mmml physnet-md --checkpoint out/ckpts/cybz_physnet --data out/splits/
energies_forces_dipoles_train.npz -o out/physnet_md
mmml physnet-md --checkpoint out/ckpts/cybz_physnet --structure molecule.xyz -o out/physnet_md --
skip-jaxmd
```

Key options:
- `--checkpoint`: PhysNet checkpoint root.
- `--data`: initial geometry from NPZ.
- `--structure`: initial geometry from XYZ/PDB/etc.
- `-o`, `--output-dir`: output directory.
- `--temperature`, `--timestep`: MD settings.
- `--nsteps-ase`, `--nsteps-jaxmd`: trajectory lengths.
- `--skip-jaxmd`: ASE-only run.
- `--n-replicas`: parallel replicas.

Typical outputs:
- `physnet_ase.traj`
- `physnet_ase_final.xyz`
- `physnet_jaxmd.xyz`

`mmml physnet-evaluate`

Evaluate a PhysNet checkpoint on an NPZ test set.

```
mmml physnet-evaluate --checkpoint out/ckpts/cybz_physnet --data out/splits/
energies_forces_dipoles_test.npz -o out/physnet_eval --plots
```

Key options:
- `--checkpoint`: checkpoint root.

- `--data`: NPZ test set.
- `-o`, `--output-dir`: metrics and predictions directory.
- `--natoms`: padded atom count if auto-detection is wrong.
- `--batch-size`, `--num-samples`, `--seed`: inference control.
- `--plots`: parity plots.
- `--subtract-atom-energies`, `--subtract-mean`: match training preprocessing.

`python -m mmml.cli.misc.train_joint`

Train the joint PhysNet+DCMNet model. This is module-level rather than a top-level `mmml` command.

```
python -m mmml.cli.misc.train_joint \
  --train-efd out/splits_dimer/energies_forces_dipoles_train.npz \
  --train-esp out/splits_dimer/grids_esp_train.npz \
  --valid-efd out/splits_dimer/energies_forces_dipoles_valid.npz \
  --valid-esp out/splits_dimer/grids_esp_valid.npz \
  --use-repo-physnet-params \
  --epochs 1 \
  --batch-size 1 \
  --name smoke_repo_physnet \
  --ckpt-dir /tmp/mmml_ckpts \
  --plot-freq 0
```

Required data:
- `--train-efd`, `--valid-efd`: E/F/D split files.
- `--train-esp`, `--valid-esp`: ESP grid split files.

Important model options:
- `--use-repo-physnet-params`: initialize PhysNet from bundled portable params.
- `--physnet-checkpoint`: initialize PhysNet from a trained checkpoint.
- `--restart`: restart a joint run.
- `--dcmnet-features`, `--dcmnet-iterations`, `--dcmnet-basis`, `--n-dcm`: DCMNet size.
- `--physnet-features`, `--physnet-iterations`, `--physnet-basis`: PhysNet size when not overridden by checkpoint config.
- `--use-noneq-model`: non-equivariant charge model instead of DCMNet.

Important loss options:
- `--energy-weight`, `--forces-weight`, `--dipole-weight`, `--esp-weight`
- `--dipole-loss-sources`, `--esp-loss-sources`
- `--loss-config`: JSON/YAML loss-term config.
- `--mix-coulomb-energy`, `--disable-physnet-point-coulomb`, `--mix-warmup-*`

Important training options:
- `--epochs`, `--batch-size`, `--optimizer`, `--learning-rate`, `--weight-decay`
- `--use-recommended-hparams`
- `--grad-clip-norm`
- `--plot-results`, `--plot-freq`, `--plot-samples`

Expected healthy startup messages:
- “Loaded PhysNet config from checkpoint”
- “Merging PhysNet params into joint model”
- “Pre-computing edge lists”

`mmml train`

Generic training entry point for DCMNet or PhysNetJAX.

```
mmml train --model dcmnet --train data/train.npz --valid data/valid.npz --output checkpoints/dcmnet
```

This command exists as a generic API, but the current tutorial uses the specialized PhysNet trainer script and `train_joint.py` for DCMNet.

`mmml evaluate`

Generic model evaluation entry point.

```
mmml evaluate --model dcmnet --checkpoint checkpoints/dcmnet --data test.npz
```

Use model-specific evaluation commands when available.

Electric-Field Model Commands

`mmml ef-train`

Train the EF equivariant model.

```
mmml ef-train \
  --train-npz out/splits_ef0_ring/energies_forces_dipoles_train.npz \
  --valid-npz out/splits_ef0_ring/energies_forces_dipoles_valid.npz \
  --test-npz out/splits_ef0_ring/energies_forces_dipoles_test.npz \
  --rot-augment --rot-perturbation 1.0 \
  --output-dir ~/ckpts/ring_ef_ptT_run2 \
  --num_iterations 2 --max_degree 2
```

Common options:

- `--train-npz`, `--valid-npz`, `--test-npz`
- `--output-dir`
- `--features`, `--num_basis_functions`, `--num_iterations`, `--max_degree`
- `--batch_size`, `--num_epochs`
- `--rot-augment`, `--rot-perturbation`
- loss weights such as `--forces_weight`, `--energy_weight`, `--dipole_weight`

`mmml ef-evaluate`

Evaluate an EF model and write metrics/plots.

```
mmml ef-evaluate --params ./ef_run/params.json --data splits/test.npz --output-dir ./ef_eval --
output-h5 ./ef_eval/eval_gui.h5
```

Use `--output-h5` when you want to inspect results in the GUI.

`mmml ef-md`

Run MD with a trained EF model.

```
mmml ef-md --params ./ef_run/params.json --data splits/train.npz --steps 5000 --output md.traj
mmml ef-md -b jax --params ./ef_run/params.json --xyz mol.xyz --thermostat langevin --steps 10000
```

Key options:

- `--backend`, `-b`: `ase` or `jax`.
- Common forwarded flags: `--params`, `--config`, `--data`, `--xyz`, `--index`, `--electric-field`, `--thermostat`, `--temperature`, `--dt`, `--steps`, `--output`.
- ASE backend supports multi-replica runs and electric-field ramps.
- JAX backend is a compiled single-replica path.

MD and Sampling Commands

`mmml run`

Run a hybrid MM/ML simulation with the PyCHARMM-integrated calculator.

```
mmml run \
  --pdbfile pdb/init-packmol.pdb \
  --checkpoint ~/ckpts/water_dimer_joint \
  --n-monomers 2 \
  --n-atoms-monomer 3 \
  --cell 25.0 \
  --ensemble nvt
```

Key options:

- `--pdbfile`: PDB to load.
- `--checkpoint`: ML checkpoint.
- `--n-monomers`, `--n-atoms-monomer`: system decomposition.
- `--cell`: cubic PBC cell length.
- `--ml-cutoff`, `--mm-switch-on`, `--mm-cutoff`: hybrid cutoff behavior.
- `--include-mm`, `--skip-ml-dimers`: toggle energy components.
- `--temperature`, `--timestep`, `--ensemble`, `--pressure`: MD settings.

- `--nsteps_ase`, `--nsteps_jaxmd`: simulation lengths.
- `--precompile`: compile JAX energy/force and exit.
- `--ml-batch-size`: chunk ML batches to reduce GPU memory.

Use this for production-style hybrid simulations; expect first-run JAX compile time to be significant.

`mmml md-system`

Run predefined mixed-composition MD setup scripts.

```
mmml md-system --setup pbc_npt --composition MEOH:5,TIP3:5 --temperature 300 --pressure 1.0
mmml md-system --setup all --n-molecules 10 --ps 1.0 --dt-fs 0.25
```

Key options:
- `--setup`: `free_nve`, `free_nvt`, `pbc_nve`, `pbc_nvt`, `pbc_npt`, or `all`.
- `--composition`: residue counts such as `MEOH:5,TIP3:5`.
- `--checkpoint`, `--output-dir`, `--template-pdb`
- `--spacing`, `--ps`, `--dt-fs`
- `--temperature`, `--pressure`
- `--extra-args -- ...`: forward raw args to the underlying script.

`mmml active-learning`

Extract useful frames from trajectories for QM relabeling.

```
mmml active-learning -i out/physnet_md/physnet_ase.traj -o md_sampled.npz --max-temp 300
mmml active-learning -i "out/*.traj" -o md_sampled.npz --stride 5 --max-frames 100
```

Key options:
- `-i`, `--input`: one or more `.traj` / `.xyz` files; globs are supported.
- `-o`, `--output`: output geometry NPZ.
- `--max-temp`: keep frames below a temperature threshold.
- `--no-temp-filter`: keep all frames.
- `--stride`: take every Nth frame.
- `--max-frames`: cap output size.

Follow with:

```
mmml pyscf-evaluate -i md_sampled.npz -o md_evaluated.npz --esp
mmml fix-and-split --efd original_train.npz md_evaluated.npz -o out/splits_extended
```

Geometry, Trajectory, and Visualization Utilities

`mmml interpolate-xyz`

Interpolate two XYZ structures in internal coordinates and write an NPZ path.

```
mmml interpolate-xyz reactants.xyz products.xyz -o path.npz --steps 500
```

Requirements:
- Same atoms and same atom ordering in both XYZ files.
- First structure defines the Z-matrix topology.

`mmml unwrap-traj`

Unwrap periodic trajectories and write trajectory/XYZ/extxyz output.

```
mmml unwrap-traj wrapped.traj -o unwrapped.extxyz --format extxyz --fast
mmml unwrap-traj traj.h5 -o unwrapped.xyz --reference frame0.xyz --group-size 3
```

Key options:
- positional `input`: trajectory, XYZ/extxyz, or HDF5 file.
- `-o`, `--output`: output path.
- `--format`: `auto`, `traj`, `xyz`, `extxyz`.
- `--fast`: streaming writer for XYZ/extxyz.
- `--cell`: override cell.
- `--group-size`, `--n-groups`: molecule-wise unwrapping.
- HDF5 helpers: `--reference`, `--coord-key`, `--numbers-key`, `--cell-key`.

`mmml sample-diverse-xyz`

Pick diverse structures from XYZ files using SOAP-style descriptors.

```
mmml sample-diverse-xyz frames.xyz -o sampled.npz --n-samples 100
```

Use this when a long trajectory has many redundant frames.

`mmml gui`

Start the molecular viewer GUI.

```
mmml gui
mmml gui --data-dir ./trajectories
mmml gui --file simulation.npz
mmml gui --data-dir ./data --port 8080 --no-browser
```

Supported common formats:

- `.npz`
- `.traj`
- `.pdb`

Useful options:

- `--data-dir`, `-d`: browse a directory.
- `--file`, `-f`: open a single file.
- `--port`, `--host`
- `--model-params`, `--model-config`: hidden-state inspection.

DCM/MDCM and Analysis Utilities

`mmml kernel-fit`

Fit a kernel mapping geometry descriptors to MDCM charge positions.

```
mmml kernel-fit charmm_ml_comparison.h5 -o kernel_out --out-h5 kernel_eval.h5 --residue-name MEOH
```

Key options:

- positional `h5`: comparison HDF5 input.
- `--out-dir`, `-o`: output directory.
- `--out-h5`: GUI/evaluation HDF5.
- `--out-mdcm`, `--out-kmdcm`: CHARMM/MDCM output files.
- `--residue-name`
- `--optimize`: optimize charge positions before fitting.
- `--train-frames`: comma-separated training frame indices.
- `--lam`, `--sigma`: kernel ridge settings.

`mmml extract-checkpoint-metrics`

Extract and plot training metrics from Orbax checkpoints.

```
mmml extract-checkpoint-metrics ~/ckpts/water_dimer_joint -o training_metrics.png --log-loss
```

Use this after PhysNet or joint training to show loss and MAE curves.

`python -m mmml.cli.misc.compare_charmm_ml`

Compare CHARMM and ML behavior on a held-out test split.

```
python -m mmml.cli.misc.compare_charmm_ml \
  --checkpoint ~/ckpts/water_dimer_joint \
  --valid-efd out/splits_dimer/energies_forces_dipoles_test.npz \
  --valid-esp out/splits_dimer/grids_esp_test.npz \
  --pdb pdb/frame_00000.pdb \
  --n-samples 10 \
  --out-dir charmm_ml_comparison
```

This is module-level, not a top-level `mmml` command. Use it for diagnostic plots and HDF5 files that can feed `kernel-fit`.

Lower-Priority or Transitional Commands

`mmml downstream`

Runs downstream analysis utilities. Use command help to inspect the current sub-options:

```
mmml downstream --help
```

`mmml train` and `mmml evaluate`

Generic interfaces. Prefer the tutorial-specific or model-specific commands when they exist:

- PhysNet: `mmml physnet-evaluate`, `mmml physnet-md`
- EF: `mmml ef-train`, `mmml ef-evaluate`, `mmml ef-md`
- Joint DCMNet: `python -m mmml.cli.misc.train_joint`

Troubleshooting Quick Hits

`JAX` / CUDA fails before the CLI starts

For CPU smoke tests:

```
JAX_PLATFORMS=cpu mmml <command> ...
```

or:

```
JAX_PLATFORMS=cpu uv run python -m mmml.cli.misc.train_joint ...
```

Output already exists

Many data-generation commands refuse to overwrite existing outputs. Choose a new `--output` / `--output-dir`, or remove the old files deliberately.

`pyscf-evaluate --efield -0.01,0,0` fails parsing

Use equals form for negative first components:

```
mmml pyscf-evaluate -i traj.npz -o out.npz --efield=-0.01,0,0
```

Units look wrong after splitting

Check:

- Did the input `F` store forces or gradients? Use `--flip-forces` only for gradients.
- Was energy already converted to eV upstream? Avoid double conversion.
- Did you subtract the correct atomic reference with `--atomic-ref`?
- Are ESP values in Hartree/e before training?

Joint DCMNet parameter loading fails

For the tutorial path, prefer:

```
--use-repo-physnet-params
```

Do not combine it with:

- `--physnet-checkpoint`
- `--restart`

Healthy startup should mention the loaded PhysNet config and parameter merge.